

HS 544: Disaster Risk Management

Course Code	HS 544
Course Name	Disaster Risk Management
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG Elective
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- **Understanding Disaster** (12 lecture hours)
 - Basic Terms and concepts: Structural failures, Pollution, Accidents, Hazard, Disaster, Catastrophes, Risk, Vulnerability, Resilience
 - Types of disaster:
 - * Natural disasters: landslide, avalanche, volcanic activities, drought, flood, cyclone, tsunami, earthquake
 - * Anthropogenic disasters: industrial pollution, biological and nuclear disaster, forest fire, oil spills, gas leak, radiations, household waste
 - * Exposure to disaster and abilities to avert risks; and, Resilience Global Conventions and National Frameworks
 - * Inventories: Maps, Zonation, Scale, Frequency, Return period risk framework Environmental risks and Climate Change: Global Conventions; National Frameworks
- **Unit II: Risk Assessment** (8 lecture hours)
 - Hazard analysis: Hazard, vulnerability, susceptibility determination Consequence analysis: categories, elements of risk, vulnerability
 - Risk: Analysis, estimation, assessment
- **Unit III: Risk Reduction Approach** (5 lecture hours)
 - Structural measures Non-structural measures Total risk Acceptable risk Cost-Benefit Analysis
- **Unit IV: Response and Disaster Management** (10 lecture hours)
 - Social Response: household, community, administrative
 - Technical Responses and Management of Disaster
 - International Conventions and Institutional and Legal Responses:
 - * Disaster Management Act 2005/ Plan 2016/2019
 - * National Policy on Disaster Management
 - * National Guidelines and Plans on Disaster Management, UNDP, UNDRR, UNSDR guidelines
- **Unit V: Sendai Risk Framework** (7 lecture hours)
 - Precursors, Framework content, Actions and Schemes,
 - Risk Governance,
 - Recovery, Rehabilitate and Reconstruction (R-R-R)
- Recent case studies from community level to national level disaster, mini projects (out of class assignments and activities)

Text Books

1. Ulrich Ranke, **Natural Disaster Risk Management: Geosciences and Social Responsibility**, Springer International Publishing, 2016.
2. R Subramanian, **Disaster Management**, Vikas Publishing house pvt., ltd., 2018

References

1. Pandey, Mrinalini, **Disaster Management**, Wiley India, 2014.
2. Ulrich Beck, **Risk Society: Towards a New Modernity**, Trans. by Mark Ritter, Sage Publications, 1992.
3. **Sendai Framework for Disaster Risk Reduction 2015-2030**, UNDRR
4. **Recently published reports - UNDRR**
5. **National Disaster Management Guidelines- 2019**, NIDM, NDMA, India
6. **National Disaster Management Policy**, 2009, GoI (As the changes in disaster management act are drastic, the above references need to be followed as recently published material)